n8n Automation Engineering - Assessment Project 5

Customer Support Ticket Triage, Routing & Auto-Responder

Project Understanding: What Needs to be Built Using the System

1. The Business Problem & Automation Goal

Customer service departments handle hundreds of incoming user inquiries, ranging from trivial password resets and product inquiries to high-impact production outages and payment failures. When tickets arrive through an unmanaged generic support channel, urgent issues get buried in standard queues, causing severe SLA breaches, client frustration, and potential revenue loss.

The goal of this project is to build an intelligent, real-time **Customer Support Ticket Triage and Routing Engine** in n8n. The workflow ingests support ticket submissions via webhooks, sanitizes and standardizes text inputs, generates a persistent ticket reference number, classifies the issue by priority (`Critical`, `High`, `Standard`) using a multi-way `Switch Node`, records the ticket into a centralized Google Sheets master ledger, and triggers tailored dual-party communications: an immediate auto-responder acknowledgment to the customer quoting their SLA target, and an urgent escalation alert to the on-duty support lead for critical tickets.

2. System Components & Architecture

- **Intake Gateway (Webhook Node):** Ingests incoming ticket payloads (`customer_name`, `email`, `category`, `priority`, `subject`, `description`) via HTTP `POST`.
- **Data Sanitization & Reference Generator (Edit Fields / Set Node):**
  - Trims whitespace from subjects and descriptions.
  - Formats emails to lowercase.
  - Generates an official tracking reference: `TCK-YYYYMMDD-XXXX`.
- **Centralized Data Persistence (Google Sheets Node):** Inserts the sanitized ticket into `Support_Ticket_Register` with initial status `Open`.
- **Priority Routing Engine (Switch Node):** Routes execution based on `{{ $json.priority }}`:
  - **Critical:** Immediate 2-hour SLA target. Triggers urgent staff escalation.
  - **High:** 4-hour SLA target. Triggers queue notification.
  - **Standard (Default):** 24-hour SLA target. Standard processing.
- **Customer Auto-Responder (Gmail Node):** Dispatches a professional HTML confirmation email to the customer containing their unique `Ticket_ID`, issue recap, and promised turnaround time.
- **Internal Escalation Alert (Gmail Node):** For `Critical` and `High` tickets, dispatches an urgent escalation notification to `support-leads@company.com`.

3. Architecture & Data Flow Diagram

```
[ Customer / Web Support Portal ]
         | (HTTP POST Ticket JSON)
         v
```

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```
[ Webhook Node: /v1/support-ticket ]
         |
         v
[ Edit Fields: Data Sanitization ] --> Generate Ticket_ID (TCK-xxxx), Trim Text
         |
         v
[ Google Sheets: Append Row ] --> Insert row in Support_Ticket_Register (Status: "Open")
         |
         v
[ Switch Node: Priority Routing ]
   +-- Case 1: "Critical" --> [ Set: SLA 2 Hours ] --> [ Gmail: Urgent Alert to Lead ]
   +-- Case 2: "High"     --> [ Set: SLA 4 Hours ] --> [ Gmail: High-Priority Queue ]
   +-- Default: "Standard"--> [ Set: SLA 24 Hours ]
         |
         v
[ Gmail: Customer Acknowledgment ] --> Send branded email with Ticket_ID & Target SLA
         |
         v
[ Respond to Webhook Node ] --> HTTP 200 {"status": "created", "ticket_id": "TCK-..."}
```

Assessment Governance & Execution Charter

Key Assessment Parameters

Parameter	Specification
Project Code	**N8N-PRJ-05**
Allocated Timeframe	**2.0 Hours** (Time-boxed practical assessment)
Lead Assessor & Reviewer	**Deepesh Jain** (Course Lead / Lead Evaluator)
Passing Score Threshold	Minimum **70 / 100 points** (100% compliance on Zero-Tolerance Gates)
Submission Format	Exported workflow `.json`, Google Sheets link, test execution matrix, and walkthrough video

1. What Gets Assessed

- Webhook configuration and immediate JSON response handling.
- String manipulation and reference ID generation in Edit Fields.
- Multi-path priority routing using the Switch node.
- Google Sheets schema definition and row append mechanics.
- Personalized HTML email creation for both customer and internal escalation team.
- Verification of distinct SLA injection based on ticket priority.

Mandatory Submission Package Specifications

1. **n8n Workflow JSON File (`.json`):** Exported workflow file with clear node hierarchy and annotations.

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2. **Supporting Documentation & Test Execution Matrix:** Google Sheet with recorded tickets across all 3 priority levels, plus test payload logs.
3. **Recorded Walkthrough Video (5-7 minutes):** Screen recording showing:
   - Webhook testing via curl / Postman.
   - Switch node evaluating Critical vs Standard priorities.
   - Verification of live rows in Google Sheets and receipt of both customer auto-reply and lead escalation emails.

2. Official 100-Point Scoring Rubric

Evaluation Dimension	Points	Scoring Criteria
Functional Delivery	**35**	Webhook intake, data sanitization, Google Sheets insert, Switch routing, and dynamic dual Gmail notifications work without fault.
Switch Routing & Logic	**20**	Multi-way Switch correctly evaluates priorities; SLA times are dynamically set without hardcoded email templates.
Data Mapping & Expressions	**15**	Proper Ticket ID formatting, date timestamping, and text trimming using n8n expressions.
Google Sheets & Gmail Integration	**15**	Accurate column mapping in spreadsheet; polished HTML email layouts with clean styling.
Testing Evidence & Traceability	**10**	Documented test runs (TC-001 to TC-003) with verifiable n8n execution IDs.
Code Hygiene & Oral Defense	**5**	Clean visual arrangement, descriptive node names, and clear explanation during evaluator review.
Total	**100**	**Passing Score: 70.0 / 100.0**

Zero-Tolerance Quality Gates (Immediate Auto-Fail)

- **Gate 1: Dropped Critical Alerts:** Failing to trigger the internal lead escalation email when a `Critical` ticket is submitted.
- **Gate 2: Exposed Internal Credentials:** Exposing Gmail app passwords or personal OAuth secrets inside action bodies.
- **Gate 3: Blocked Webhook Endpoints:** Failing to return an HTTP response to the webhook caller, causing frontend client timeouts.

Project Requirements Specification

1. Document Control

Field	Value
Document Title	Requirement Document: Customer Support Ticket Triage, Routing & Auto-Responder
Course	n8n Hands-On Automation Course (Topics 1-15)
Author / Reviewer	**Deepesh Jain** (Course Lead / Lead Evaluator)

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Timeframe	**2.0 Hours**
Status	Approved for Learner Assessment

2. Business Requirements Document (BRD)

- **BRD-001 (Real-Time Ingestion):** Customer support requests must be captured immediately upon form submission.
- **BRD-002 (Priority Classification):** Tickets must be triaged into Critical, High, or Standard priorities to meet corporate SLAs.
- **BRD-003 (Customer Transparency):** Requesters must receive instant confirmation quoting their ticket ID and expected response time.
- **BRD-004 (Internal Escalation):** Critical incidents must instantly alert the senior engineering lead on duty.

3. Product Requirements Document (PRD)

- **PRD-001:** Webhook node listening on `/v1/support-ticket` for `POST` requests.
- **PRD-002:** Edit Fields node generating `TCK-YYYYMMDD-XXXX` IDs.
- **PRD-003:** Google Sheets node appending row to `Support_Ticket_Register`.
- **PRD-004:** Switch node routing based on `priority`.
- **PRD-005:** Gmail nodes dispatching customer confirmation and internal alerts.

4. Software Requirements Specification (SRS)

4.1 Functional Requirements

- **FR-001 (Intake Payload):** Receive fields: `customer_name`, `email`, `category`, `priority`, `subject`, `description`.
- **FR-002 (Sanitization):** Generate unique ID: `='TCK-' + $now.toFormat('yyyyMMdd') + '-' + Math.floor(1000 + Math.random() * 9000)`
- **FR-003 (Persistence):** Insert new row into `Support_Ticket_Register` with status `Open`.
- **FR-004 (Priority Switching):**
  - Case 1: `Critical` > Injects `sla_hours = 2`, triggers internal alert.
  - Case 2: `High` > Injects `sla_hours = 4`, triggers internal queue notification.
  - Default: `Standard` > Injects `sla_hours = 24`.
- **FR-005 (Customer Acknowledgment):** Dispatch HTML email to `{{ $json.email }}` quoting `{{ $json.Ticket_ID }}` and `{{ $json.sla_hours }} hours`.
- **FR-006 (Lead Escalation):** If `Critical`, dispatch email to `support-leads@company.com` with subject: `[ESCALATION] Critical Support Ticket: {{ $json.Ticket_ID }}`.

4.2 Non-Functional Requirements

- **NFR-001 (Response Latency):** Webhook must return HTTP 200 to the submitter within 2,000 ms.

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- **NFR-002 (Case Insensitivity):** Priority matching in Switch must handle lowercase and uppercase (`critical`, `CRITICAL`).

Technical Specifications & Integration Details

1. Google Sheets Data Schema (`Support_Ticket_Register`)

Column Name	Data Type	Required	Sample Value	Description
`Ticket_ID`	String	Yes	`TCK-20261016-5120`	Autogenerated unique reference
`Created_At`	DateTime	Yes	`2026-10-16 14:40:00`	Ingestion timestamp
`Customer_Name`	String	Yes	`Johnathan Doe`	Requester name
`Customer_Email`	String	Yes	`j.doe@enterprise.org`	Requester contact
`Category`	String	Yes	`Infrastructure`	Inbound category
`Priority`	String	Yes	`Critical`	Triage priority
`Subject`	String	Yes	`Production Database Outage`	Issue title
`Target_SLA_Hrs`	Number	Yes	`2`	Resolved SLA turnaround
`Status`	String	Yes	`Open`	Lifecycle state

2. Sample Inbound Webhook Payload

```
{
  "customer_name": "Johnathan Doe",
  "email": "j.doe@enterprise.org",
  "category": "Infrastructure",
  "priority": "Critical",
  "subject": "Production Database Connection Timeout",
  "description": "Our primary production database cluster is not accepting new SSL connections since 14:00."
}
```

3. Node Expression Formulas

- **Ticket ID Generation:**

```
='TCK-' + $now.toFormat('yyyyMMdd') + '-' + Math.floor(1000 + Math.random() * 9000)
```

- **Switch Condition Expression:**

```
={{ $json.priority.trim().toLowerCase() }}
```

Test Execution Matrix

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Test ID	Test Scenario	Input Data	Expected Outcome	Execution Evidence Required
TC-001	**Critical Ticket Submission**	Priority: `Critical`, Category: `Infrastructure`	Row saved in sheet; Customer gets 2-hr SLA email; Lead gets urgent alert.	Screenshot of Google Sheet row + 2 Gmail messages
TC-002	**Standard Ticket Submission**	Priority: `Standard`, Category: `Billing`	Row saved in sheet; Customer gets 24-hr SLA email; Lead alert is skipped.	Screenshot of Google Sheet row + 1 customer Gmail message
TC-003	**Webhook Immediate Response**	Postman POST to `/v1/support-ticket`	Returns HTTP 200 with JSON body `{"status": "created", "ticket_id": "TCK-..."}`.	Screenshot of Postman HTTP 200 response

Candidate Pre-Submission Checklist

- [ ] [ ]
  Workflow is exported as `N8N_PRJ_05_Ticket_Triage.json`.
- [ ] [ ]
  Webhook node returns HTTP 200 response upon submission.
- [ ] [ ]
  Switch node correctly differentiates between Critical, High, and Standard.
- [ ] [ ]
  Google Sheet schema matches table columns exactly.
- [ ] [ ]
  5-minute video walkthrough demonstrates submission and dual email delivery.

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